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MARICO LTD.: DISTRIBUTION NETWORK OPTIMIZATION

Aishwarya Sadasivan, Ashish Pingulkar, and Debmallya Chatterjee wrote this case solely to provide material for class discussion. The authors do not intend to illustrate either effective or ineffective handling of a managerial situation. The authors may have disguised certain names and other identifying information to protect confidentiality.

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On July 1, 2020, Arun Parekh, the co-head of logistics and planning at Marico Ltd. (Marico), was planning a well-deserved weekend after a hectic week of work. He had worked with the company for several years and had been promoted to his current position eighteen months earlier. He was about to leave work that day when he received an urgent notification for a virtual meeting from Naren Shah, his colleague and co-head of the department.

Brushing past formalities, Naren came straight to the point. He screen-shared a Microsoft Excel file with figures on the company's warehouses that showed high inventories and low service levels. In a meeting scheduled with the leadership team the following week, Naren and Arun would have to explain these figures and recommend a strategy for the future. COVID-19 pandemic lockdowns imposed by state and national governments had severally impacted supply chains[1], but why were service levels low in warehouses with more than adequate inventory? Naren felt that consolidating warehouses would help address the issue, but Arun did not favour employee cutbacks, which could further exacerbate the issue of service levels.

Despite the pandemic, Marico's revenue levels had remained largely unaffected, even seeing improvements in some territories. However, some warehouses were reporting high inventory with low service levels, while the logistics and planning division was pressured to reduce costs and show a profit in the September quarterly reports. Arun was doubtful about Naren's solution. He asked Alia Gill, his talented intern, to review the company's distribution design and recommend an appropriate plan to address these issues. Arun would then present Alia's findings at the following week's leadership meeting. Alia had a demanding task ahead of her. She would have to review and assess the company's current situation and overall business model, and then prepare a set of recommendations.

THE COMPANY

Marico was a fast-moving consumer goods conglomerate that operated eight different plants across India, with a footprint in twenty-five other countries in Asia and Africa. The company generated annual revenues of over

[1] Nistula Hebbar, "PM Modi Announces 21-Day Lockdown as COVID-19 Toll Touches 12," *The Hindu*, December 3, 2021, https://www.thehindu.com/news/national/pm-announces-21-day-lockdown-as-covid-19-toll-touches-10/article31156691.ece.

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₹70 billion in 2019–20[2] selling a variety of products including value-added hair oil,[3] coconut natural oil, and edible oil, which collectively accounted for over 90 per cent of the company's revenues. Marico's coconut natural oil product line included the popular brands Parachute and Coco Soul, which was produced from cold pressed pure coconut oil. The company also sold scented versions of coconut oil, such as jasmine and aloe vera, under brand names such as Nihar Shanti and Parachute Advanced, which were exclusively meant for hair care. The company's only edible oil product, Saffola, came in four variants: active, gold, tasty, and total.

Coconut oil manufacturing facilities were situated near the source of the raw material, which made inbound transportation efficient and cost effective. At two Tamil Nadu facilities, in Perundurai and Pondicherry, the dried coconut substance copra was expelled to extract coconut oil. The blending of edible oils was done in the Maharashtra city of Jalgaon. The process involved blending together different oils, such as sunflower and rice bran, which were also extracted near the source of origin of the raw material. After the blending process was completed, the edible oil was packaged for shipping. Consumers in the western region of India, especially in the state of Maharashtra, consumed large amounts of edible oil. Marico set up a plant in this area to address high demand, which could vary in different periods, and to ensure product freshness.

Marico's value-added hair oil plants in Guwuhati (in the state of Assam) and Baddi (in the state of Himachal Pradesh) catered to consumer demand in the northern part of the country, while the company's plant in Sanand (in the state of Gujarat) fulfilled demand in the south. These plants were strategically located to address category and regional-specific demand for Marico's products. Product distribution took place through twenty-five warehouses located across the country. Marico's distribution network followed a three-echelon model of plant, warehouse, and distributor.[4]

CURRENT DISTRIBUTION

All orders from distributors were shipped from Marico's twenty-five warehouses. At one point, the company had as many as thirty-four warehouses—one in each Indian state. The size, location, and capacity of each warehouse varied according to the nature and volume of product demand in that region. The company reduced the number of warehouses down to twenty-five after the introduction of India's Goods and Service Tax in 2017.[5] The government's new tax legislation eliminated the cascading effect of taxes, particularly interstate taxes. However, three years later, Arun and Naren were once again reviewing the company's distribution network with the goal of controlling high inventory and low service levels. Naren recommended consolidating warehouses, but Arun wanted to explore other alternatives.

ELEMENTS OF THE CURRENT NETWORK DESIGN

Abandoning all weekend holiday plans, Arun set to work early the next morning to delineate his strategy by making notes about the factors that needed attention. He started with the two main factors that determined product distribution: demand and freight rates.

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[2] ₹ = INR = Indian rupee; US1$ = ₹75.5 on July 1, 2020; all currency amounts are in ₹ unless otherwise specified; Marico Ltd., *Integrated Report 2019–20*, accessed March 15, 2021, https://marico.com/investorspdf/Annual-Report-FY-2019-20.pdf.

[3] Value added hair oil products, which were intended for hair nourishment, included the brands Nihar Shanti Amla, Parachute Advanced, and Hair & Care; coconut natural oil products, which were used for both cooking and hair nourishment, were made purely from coconut oil with no additives and included the brand Parachute; Marico Ltd., *Integrated Report 2019–20*.

[4] Makoena Sebatjane and Olufemi Adetunji, "A Three-Echelon Supply Chain for Economic Growing Quantity Model with Price- and Freshness-Dependent Demand: Pricing, Ordering and Shipment Decisions," *Operations Research Perspectives* 7 (2020): 1, https://www.sciencedirect.com/science/article/pii/S2214716020300439?via%3Dihub.

[5] Vikas Vasal and Armaan Joshi, "What Is GST? 5 Things You Need to Know," *Forbes*, July 27, 2021, https://www.forbes.com/advisor/in/tax/what-is-gst-5-things-you-need-to-know.

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Regarding demand, Arun wondered if he should widen his focus beyond individual districts. He immediately contacted his team member Manoj and asked him to compile the sales numbers of various districts.

Regarding fright rates, Arun asked Manoj to prepare accounts divisions for final freight rates from all plant and warehouse routes to the distributors, for the previous eight or nine months. Arun knew that freight rates were determined by various different factors, including destination, load, and distance travelled. Data insights from these factors were used to negotiate rates with the transport agencies.

By that same afternoon, Arun had all data in hand. Before taking his lunch break, he asked Alia to update him on the status of the data he had asked for earlier. Alia was reviewing the company's distribution design before making recommendations for an appropriate plan of action. Accordingly, Alia approached the issue using a concept she had learned in her network designing course, and was eager to find a solution.

However, there were two contentious factors to consider, as Arun pointed out to Alia. First, location data was available for the general trade distributors, but not for modern trade and e-commerce. The distribution network was set up to cater to distributors of the general trade, who were higher in number and more widely scattered, but modern trade and e-commerce were managed from the nearest general trade warehouse. Second, because Marico used roadways for logistics, it should be the key focus area for planning the distribution design.

Alia duly noted the constraints with respect to the data points at hand and decided to read up on developing a model for a roadways distribution system. By that evening, Arun received an overview of the company's distribution across the country. However, this data was for about 400 districts and included thirty-three files on freights across eight plants and twenty-five warehouses over the previous eight or nine months. In the interest of time, Arun again called Alia and asked her to go through the data to identify trends in sales and freight rates and then report to him her insights and the recommended plan by the next morning.

Alia set to work and went through the deliverables agenda she had created during the day. She had managed to create an agenda based on her understanding of network design (see Exhibit 1). She was well on her way to discover the first two deliverables from the data on freight.

DATA ANALYSIS

Freight Analysis

Alia used the freight details as the starting point in the analysis and reviewed the notes she had made during her call with Arun. She noted that factors such as destination, distance, and loadability determined the freight rates. The term "loadability" referred to a measure of the truck's utilization in terms of weight. Initially, the enormity of the data seemed overwhelming. However, Alia managed to get a head start by organizing, separating, and labelling all data under different category headings. She then performed regression analysis between the specified freight rate and the loadability and distance. She used kilograms (kg) as the y variable and distance as the x variable to get an overview of the analysis (see Exhibit 2).

Given that the primary freight was a full truckload, the equations had equal variable components and the intercept varied across plants. This could be attributed to the kind of product being transported from each plant; however, the cost per kg across all lanes was constant, regardless of the destination of the load. Secondary freight was not constant; it depended on the destination. This was due to factors such as return load and order volumes. Essentially, secondary freight was not specific to the distributor's location, but it was dependent on the warehouse location (see Exhibit 3).

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Therefore, Alia divided the entire country into nine zones for her analysis (see Exhibit 4), for which the regression analysis was stabilized. These nine zones were based on destination. This was because for each destination, tariff rates were more or less fixed. These nine zones also defined the type of loadability across the country, given that demand varied and secondary freight was characterized by a less-than-full-truckload attribute. Loadability was therefore also specific to the particular destination or to demand.

Alia also found that the equations in the primary freight had an intercept value, whereas those of the secondary freight were 0 intercept equations. The intercept was proof that full truckloads had more stability in cost, irrespective of the lanes. However, in case of less-than-full truckloads, variability depended on the lane being traversed. This meant that warehouses should be closer to the high demand density areas (demand density = demand ÷ area). Also, the number of warehouses should be determined by the trade-off between service level and cost of servicing. As Alia wrote down her latest findings, she realized the time was 3:00 a.m. She wisely decided to stop her work and get to bed. She would discuss her findings with Arun the next morning.

The next day, Alia met virtually with Arun and explained how primary and secondary freight calculations differed due to the full-truckload and less-than-full-truckload features, which made the primary freight largely dependent on distance and product. Secondary freight, on the other hand, depended on the destination of the shipment. Alia also recommended to Arun that a fresh perspective on warehouse location was needed to identify existing gaps and constraints the business was operating under.

Arun was impressed with the regression analysis that Alia had conducted, although he was not sure a fresh perspective on location was necessary. Alia recommended reaching an ideal solution and accommodating real-time business constraints, while modelling the linear programming to minimize freight for a given service level. Arun approved Alia's request, but wondered if the analysis results would present a major deviation from the current situation.

Warehouse Location Identification

After her focus on freight, Alia moved on to the equally important factor of location, which would help determine various cost factors including labour and rent versus distance. She decided to review average demand figures for the previous year (i.e., 2018–19). The sales data showed that millions of records were available for that period, which would make initial data processing in Excel a tedious task. However, given that demand had stabilized over the years, she decided that the past year's data would make a reasonable benchmark for analysis.

To study demand, she consolidated the data spread over 400 districts. To further consolidate the data, she deliberated on the action of warehousing, which was essentially consolidating demand. She could correlate the action of consolidation with clustering; therefore, she decided to use hierarchical clustering to view demand. Three factors to consider were warehousing, constraints on warehouse size, and distance travelled to reach distributors. However, to understand geographical distribution, she wanted to freeze and eliminate factors such as demand size and serviceability in terms of the maximum distance covered for servicing.

Alia ensured that each cluster was equisized in terms of demand and covered distributor locations within a one-day service level diameter within a 200 kilometre (km) distance (< = 200 km). With these specifications in place, demand was represented in the form of fifty-three centroid locations (see Exhibit 5). These locations indicated the weighted average of demand and helped analyze locations independent of demand, focusing only on geography of distribution.

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The centroid represented the centre of the cluster inclining to greater demand, because these geographies were pivotal to the business and were serviced more often. Therefore, from the perspective of service level and on-time in full levels, Alia had to find a cluster representative via a weighted average methodology by giving both demand and distance equal representation.

Unlike the usual practice of considering the conventional warehousing locations for an optimization function, this approach would help raise certain non-conventional warehousing locations that were relevant to the company. Using this method, Alia was able to identify fifty-three centroid locations for Arun to consider for his presentation.

FIRST PHASE OF THE PRESENTATION

Alia's freight analysis showed that of the top ten districts in terms of demand, only eight had warehouses to cater to their demand, which strengthened Arun's argument that more warehouses were needed in new locations. The fifty-three centroid locations helped to pinpoint potential warehouse centres that would deliver one-day service across the country. But future optimization would help pick and choose important geographic areas from both demand and freight perspectives. Arun seemed satisfied with the initial analysis and provided some inputs for Alia to consider for the next phase of the presentation.

The next day at 10:00 a.m., Alia met virtually with Naren and Arun to discuss her progress on a new distribution design for the company. Specifically, she intended to identify network distribution gaps and suggest how to increase service levels across the country, while optimizing freight.

The analysis of the current top ten districts showed that two locations had no warehouse, which Naren found particularly interesting. Alia pulled up the freight equations and the centroid locations of demand. Weaving these elements together, she presented her solution, which detailed the tangible final locations of warehouses, freight, and service levels. Alia's analysis outlined the following steps (see Exhibit 1):

1) *Warehouse identification*—Identify the initial set of candidate locations for potential warehouses. These locations represented the ideal centres within the one-day service level. The idea was to choose from the multiple hotspots on the basis of demand distribution.
2) *Warehouse selection*—Use quantitative and qualitative reasoning to select the initial list of hotspots based on demand. This reasoning would be partially designed in a mathematical formulation and partially in the form of qualitative analysis. Together, these tasks would help establish the best locations for the warehouses as follows:
   a) *Design and determine the objective function*—This was the main objective of the business—to reduce freight and increase service levels. This meant that stock should be available at all geographical areas at the optimal cost.
   b) *Calculate the elements of the objective function*—These elements would include equations of primary and secondary freight, which Alia had already calculated to bridge the gap between geographical peculiarities in terms of freight rates versus the demand fulfillment requirement.
   c) *Calculate the ideal number of warehouses to optimize cost and service levels*—An increase in the number of warehouses would increase service levels and the overall freight. It was necessary to identify the point at which the trade-off between the two measures would not make sense. Essentially, the incremental benefit from increased service level was not equivalent to the increment in the overall freight.

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d) *Develop constraints*—There were business constraints for service levels and depot management. For example, a distributor would be required to deal with one specific warehouse to make sourcing easier, even if the overall freight level was higher due to the extra distance.

3) *Practical implications and alternative analysis*—Practical implications involved primarily warehouses that were pivotal to the business and should therefore not be disturbed at any cost. This implied that the design would have to be reanalyzed and reconfigured around these considerations.
4) *Qualitative analysis*—Once the quantitative aspect of freight, distance, and demand were accommodated, the next step would be to evaluate the plausible warehouse locations and their rents for the final cost implications.
5) *Final recommendation*—Conduct a trade-off analysis between the ideal and alternative design suggestions and come up with a final recommendation.

The way forward was quite specific and clear in its objective. Naren and Arun were worried about the trade-offs involved between the alternative design implications and the ideal solution. But what if the two options were disparate? Would Naren and Arun still have to make difficult choices? These questions only added pressure to come up with the right answers by the following week's leadership team meeting. The clock was ticking!

OPTIMIZATION OF THE SOLUTION

Alia got to work on the ready pieces of information on freight and initial candidate locations for warehouse locations. The next step was to form a mixed integer linear programming function for the given problem statement. Essentially, linear programming would help optimize the dilemma between fulfilling service levels criterion while reducing service cost in terms of freight. As a computer engineer, Alia was well-versed in the programming language Python and the accompanying modelling library PuLP. She used the PuLP library to design this optimization function and Excel to interpret the results.

The first step was to define the objective function for the optimization problem. The objective function for this case was to minimize the cost to serve customers, which mainly included transportation and warehousing costs. However, warehousing costs differed largely across the country and there was no way to quantify them in the form of an equation or function. Therefore, Alia kept the objective as transportation cost and evaluated the warehousing cost as a part of the qualitative analysis.

Freight was defined for every route from source to destination. Therefore, the first step in defining the objective function was to form variables of the various routes possible, from plant to warehouse and from warehouse to the distributor district.

The objective function was defined as follows (see Exhibit 6):

∑ Primary (coefficient, distance) × demand + ∑ Secondary (coefficient, distance) × demand
The freight functions in terms of "per kg per km" had previously been calculated countrywide.

Next, she defined the constraints that would limit the objective function and define an optimal point for it to minimize. She also needed certain logical constraints to achieve a realistic output. For example, the total inflow of goods had to be equal to the total outflow from the warehouse.

Constraint (see Exhibit 7):

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- Marico had an average service level promise of two days. This implied that no warehouse could be farther than 500 km from the distributor. This was a constraint because it was not possible for all warehouses to perfectly match that condition. The equation, which could include an average distance of 500 km to maintain an average of two days, was as follows:

$$\text{distance} \sum \sum \text{warehouse}_a \text{demand}_b \div (\text{number of demand points serviced}) <= 500$$

- The logical constraint to prevent illogical allocation of shipments was as follows: ∑ inflow at warehouses = ∑ outflow from warehouses
- The logical constraint to prevent illogical allocation of shipments was as follows: ∑ outflow from plants = ∑ inflow at warehouses

While the objective function and constraints were the main elements of an optimization problem, the ideal cut-off criteria also had to be defined to ensure that the optimization function provided the best minimum cost to serve. Using the graphical approach to calculate freight, Alia could determine the number of warehouses.

A higher service level required lower lead times, which could be achieved by locating warehouses closer to demand. However, this would mean setting up a greater number of warehouses, which would decrease the overall freight but also add more warehousing cost to the equation. Therefore, a trade-off between the increasing service level and reducing incremental benefit of freight would help establish the ideal position for the business to operate (see Exhibit 8). This was defined as the point where incremental benefit in service level exceeded the incremental reduction in freight, which implied that the additional warehouse only marginally improved cost, while significantly increasing the service level (see Exhibit 9).

Alia iteratively calculated the freight and service levels for ten levels of warehouses (from twenty to thirty warehouses). Using a graphical approach, she plotted the freight and service levels along the simultaneous axes and found the ideal number of warehouses at the point of intersection along the two graphs. The freight was reduced with an increase in warehouses, while the service levels increased with an increase in warehouses. Therefore, the two opposing graphs intersected at the ideal point.

PRACTICAL CONSTRAINTS AND ALTERNATIVE ANALYSIS

After Alia had identified the ideal cut-off point, she applied it as a constraint to the optimization problem to come up with an outcome. However, she found that several critical locations for the business were missing, mainly due to the following two issues:

- Certain districts had a significant contribution of modern trade and e-commerce, where general trade business was less than e-commerce plus modern trade business.
- The unavailability of exact locations of these modern trade and e-commerce businesses made it difficult to calculate distances and freight, making it difficult to capture these instances in the system.

Alia had to accommodate the above two issues by constraining the model to force open the existing warehouses servicing the locations with greater modern trade and e-commerce demand. Due to lack of data to plot these figures in the current optimization problem, she assumed that the current locations were necessary to fulfill the service levels set by the company. With this assumption, Alia got her final set of twenty-four warehouses and was satisfied with the outcome. It satisfied the twin requirements of reducing freight and increasing service levels.

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However, Alia then realized there were two other issues with the optimized outcome:

- *Practical implications*—Shutting and opening new warehouses at six locations was an expensive and cumbersome task because the company would have to rent new spaces, hire new workers, and close down some existing spaces, in addition to paying for shipping the remaining inventory from the old warehouses to the newer locations.
- *Local optima issue*—The twenty-four locations suggested could be a "local optima," which meant the locations were ideal for the local area but created an unstable overall configuration. This problem was created by components of the optimization function being non-linear in nature. More locations could be closed down by ensuring that only the most essential locations were kept in place (see Exhibit 10).

Therefore, Alia decided to evaluate an alternative analysis to reduce some of the new locations by either mapping them to existing ones or consolidating the new ones. She observed the following possibilities:

- When a given location was forced to shut down, an alternative location would be opened in the same state within a distance of 400 km. In such a case, the affected geographical location required support, due to the demand quantum. The change would cause a severe drop in service levels and an increase in freight.
- When a given location was forced to shut down, an alternative location would be opened in another state within a distance of 1,000 km. This option indicated a lower desirability and need for the specific location. Lower desirability was suggested by a light drop in service levels but an increase in overall freight.

Alia identified twenty-two locations that were suboptimal in terms of freight and service levels, compared to the twenty-four ideal locations suggested in the previous iteration. She also noticed that the solution suggested by the optimizer with a hard constraint of twenty-two warehouses had a lower service level, compared to the alternative twenty-two warehouses solution. The possible reason could be that the optimizer was considering only the freight factor, whereas a combination of the two factors was required.

Although the company's focus was on reducing costs, Marico's primary business was to fulfill demand. Therefore, service levels were an integral component of the business objective. Finding the ideal configuration of twenty-four warehouses would help maintain service levels to a great extent, while reducing costs within the possible constraints of the business. With these considerations in place, there were several other factors to review, including rental expenses and labour costs. However, this was a more managerial decision than a technical issue.

The model could only answer questions up to a point; the final decision had to be made while considering all relevant factors. Alia would have to make a separate qualitative analysis to help management make its decision. She quickly pulled up an Excel spreadsheet and outlined a comparison of the two situations to present to Arun.

ARUN'S PROBLEM

Alia presented her final analysis to Arun, who took some time to decide which was more suited to the current scenario, future practical implications, and overall trade-offs between cost and benefit. The first option was to have twenty-four warehouses across the country. This seemed like the most optimal solution from a service level and freight perspective.

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As Arun was aware, the major cost in distribution was warehouse rent, which varied across the country. With an approximate rental rate of ₹12 per square foot (0.09 square metres) and an average size of 30,000 square feet (2,800 square metres) for a warehouse, the total warehousing monthly rental cost was ₹9,360,000 for twenty-four warehouses or ₹8,580,000 for twenty-two warehouses.

Given that freight costs of twenty-two warehouses exceeded that of twenty-four warehouses by ₹100,000 per month, Arun made his own calculations. Freight and warehousing together comprised 80–85 per cent of distribution costs, which was critical for the analysis. Apart from these cost considerations, he had to also consider less specific expenses, such as remapping distributors to newer warehouses, adjusting the number of employees to accommodate shifts, and relocating distributors according to credit periods and financial implications.

Did these calculations mean that twenty-two warehouses were a better financial plan than twenty-four warehouses? Should growth in demand also be considered? What elasticity of demand could twenty-two warehouses accommodate? Arun could see an improvement in service levels either way, considering the current state of distribution (see Exhibits 11 and 12). This would surely be a strong point for the presentation to the leadership team. However, now that rough calculations of freight and warehousing were available, should Arun choose twenty-two or twenty-four warehouses?

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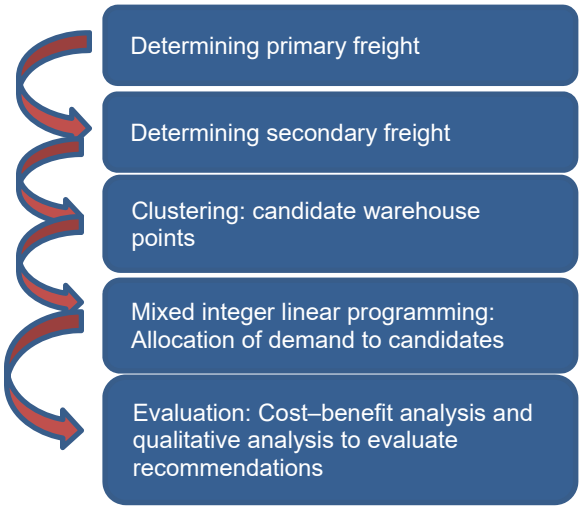
EXHIBIT 1: QUANTITATIVE ANALYSIS STEPS FOR WAREHOUSE OPTIMIZATION



Source: Created by the authors.

EXHIBIT 2: PRIMARY FREIGHT EQUATIONS

Plant	Guwuhati	Dehradun	Baddi	Jalgaon	Kanjikode	Perundara	Pondicherry	Sanand
Intercept (₹/kg)	2.73	3.84	3.84	2.52	1.64	1.76	2.1	2.85
Slope (slope) (₹/km/kg)	0.0020	0.0052	0.0052	0.0048	0.0042	0.0043	0.0044	0.0050

The above intercept and x variable represent the linear equation of cost per kg for primary freight: $y = mx + c$.
y = total cost (₹) per kg
m = cost (₹) per kg per km
x = distance (km)
c = intercept or mean value of freight represented as ₹ per kg

Note: ₹ = INR = Indian rupee; kg = kilogram; km = kilometre
Source: Created by the authors based on facts presented in the text.

EXHIBIT 3: SECONDARY FREIGHT EQUATIONS

Region	Equation
North East	= 0.0145x
North	= 0.011875x
West	= 0.0115x
South West	= 0.010625x
South East	= 0.009125x
Central	= 0.008875x
East	= 0.008x
North Central	= 0.007875x
North West	= 0.006875x

Here the output is in the form of $y = mx$.
y = total cost (₹)/kg for secondary freight
m = cost (₹)/kg/km
x = distance (km)

Note: ₹ = INR = Indian rupee; kg = kilogram; km = kilometre; x = times
Source: Created by the authors based on facts presented in the text.

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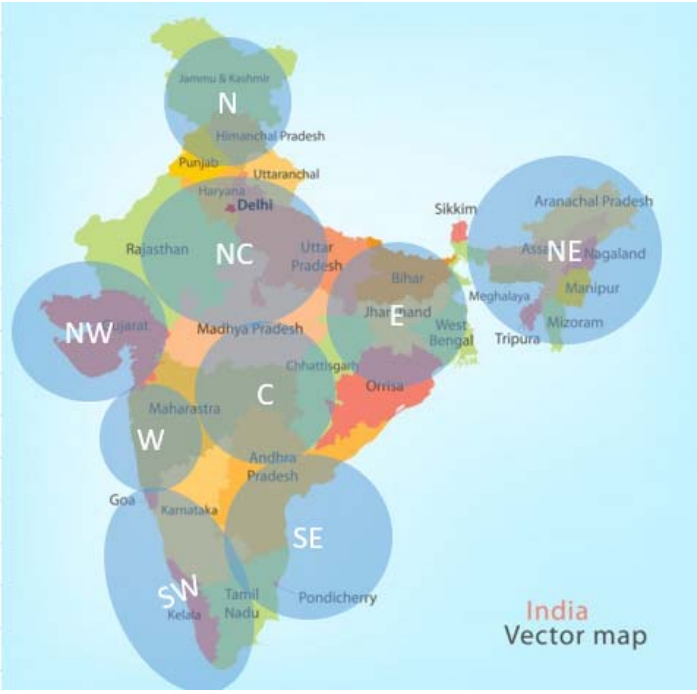
EXHIBIT 4: NINE ZONES FOR SECONDARY FREIGHT CALCULATIONS



Note: N = north; C = central; E = east; W = west; S = south
Source: Created by the authors based on facts presented in the text.

EXHIBIT 5: CENTROID LOCATIONS (53) NEEDED FOR ONE-DAY SERVICE OF ENTIRE COUNTRY

Warehouse	Latitude	Longitude	City	State
W1	27.59137	76.43122	Roondh Rampur	Rajasthan
W2	26.85383	84.19225	Amwas Khas	Uttar Pradesh
W3	17.20569	81.53633	Rajahmundry	Andhra Pradesh
W4	28.56098	80.49032	Sumer Nagar	Uttar Pradesh
W5	29.25725	77.91551	Bhoomma	Uttar Pradesh
W6	31.91242	76.05651	Kasba	Himachal Pradesh
W7	13.02118	78.39280	Gennerahalli	Karnataka
W8	20.60115	76.26120	Taroda	Maharashtra
W9	23.03439	88.81676	Bangaon	West Bengal
W10	27.33021	95.48732	Lakhipathar	Assam
W11	21.86046	73.78392	Panchmuli	Gujarat
W12	16.27019	75.24948	Bagalkot	Karnataka
W13	17.82108	78.88491	Bhongir	Telangana
W14	26.55461	81.66674	Para	Uttar Pradesh
W15	22.37411	83.58641	Raigarh	Chhattisgarh
W16	23.52274	73.49236	Behdaj	Gujarat
W17	9.607880	77.85228	Moolippatti	Tamil Nadu
W18	12.62172	76.88276	K. Gowdagere	Karnataka
W19	27.47615	79.01114	Khiriya Nagar Shah	Uttar Pradesh
W20	12.52546	79.76196	Kancheepuram	Tamil Nadu
W21	20.41729	86.45314	Parakula	Odisha

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EXHIBIT 2 (CONTINUED)

Warehouse	Latitude	Longitude	City	State
W22	24.75631	78.74415	Gangasagar	Uttar Pradesh
W23	24.09555	87.79077	Karkaria	West Bengal
W24	25.83937	86.54676	Parari	Bihar
W25	23.28316	80.48695	Sakari	Madhya Pradesh
W26	23.19029	71.02871	Kutch	Gujarat
W27	24.82048	93.59155	Tamenglong	Manipur
W28	15.69152	78.36632	Nandyal	Andhra Pradesh
W29	14.54547	79.55849	Bandarupalli,	Andhra Pradesh
W30	19.55980	73.63010	Hinglud	Maharashtra
W31	25.35598	76.39305	Tisaya	Rajasthan
W32	11.67809	92.71646	Port Blair	Andaman and Nicobar
W33	17.70758	82.50955	Anakapalle	Andhra Pradesh
W34	22.53180	77.09682	Dewas	Madhya Pradesh
W35	34.06834	74.79684	Batmaloo	Srinagar
W36	21.32824	82.51482	Dewgaon	Chhattisgarh
W37	23.26714	78.21346	Raisen	Madhya Pradesh
W38	24.04213	82.42423	Suhira	Madhya Pradesh
W39	27.66410	74.46837	Nagaur	Rajasthan
W40	25.39387	84.01259	Kasimpur	Bihar
W41	23.08160	86.36763	Rupapatia	West Bengal
W42	18.64162	77.13884	Wanjarwada	Maharashtra
W43	11.22730	76.97794	Coimbatore	Tamil Nadu
W44	18.83478	74.83712	Mandavgan	Maharashtra
W45	10.59460	79.75781	Venmanacheri	Tamil Nadu
W46	21.36683	80.17494	Purgaon	Maharashtra
W47	26.56603	93.07447	Deusur Sang	Assam
W48	14.32428	76.49978	Chitradurga	Karnataka
W49	26.73609	91.24404	Baksa	Assam
W50	20.70395	78.64692	Wardha	Maharashtra
W51	25.40305	73.72323	Kaklawas	Rajasthan
W52	31.16756	77.65145	Shimla	Himachal Pradesh
W53	23.05970	76.57799	Kalyanpura	Madhya Pradesh

Note: W = warehouse
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EXHIBIT 6: DEFINING OBJECTIVE FUNCTION AND DECISION VARIABLES FOR MIXED INTEGER LINEAR PROGRAMMING

```
#create decision variables
route_decision_cno=LpVariable.dicts("Route_CNO",(var_warehouse,var_demand),0,1,cat='Integer')
route_decision_edible=LpVariable.dicts("Route_Edible",(var_warehouse,var_demand),0,1,cat='Integer')
route_decision_vaho=LpVariable.dicts("Route_VAHO",(var_warehouse,var_demand),0,1,cat='Integer')

route_decision_p2w_cno=LpVariable.dicts("Route_P2W",(plant_cno,var_warehouse),0,1,cat='Integer')
route_decision_p2w_edible=LpVariable.dicts("Route_P2W",(plant_edible,var_warehouse),0,1,cat='Integer')
route_decision_p2w_vaho=LpVariable.dicts("Route_P2W",(plant_vaho,var_warehouse),0,1,cat='Integer')

demand_decision_cno=LpVariable.dicts("Demand_CNO",(var_warehouse,var_demand),0,None,cat='Integer')
demand_decision_edible=LpVariable.dicts("Demand_Edible",(var_warehouse,var_demand),0,None,cat='Integer')
demand_decision_vaho=LpVariable.dicts("Demand_VAHO",(var_warehouse,var_demand),0,None,cat='Integer')

demand_decision_p2w_cno = LpVariable.dicts("Demand_P2W",(plant_cno,var_warehouse),0,None,cat='Integer')
demand_decision_p2w_edible = LpVariable.dicts("Demand_P2W",(plant_edible,var_warehouse),0,None,cat='Integer')
demand_decision_p2w_vaho = LpVariable.dicts("Demand_P2W",(plant_vaho,var_warehouse),0,None,cat='Integer')


warehouse_open= LpVariable.dicts("Warehouse_Func",var_warehouse,0,1,cat='Integer')
```

Decision Variables

Objective Function

```
#define objective
model_t1 += lpSum([demand_decision_p2w_cno[p][w]*plant_w[(p,w)] for (p,w) in route_p2w_cno])+lpSum([demand_decision_cno[w][d]*dis
route_p2w_vaho])+lpSum([demand_decision_vaho[w][d]*distance_w2d[(w,d)]*secondary_coeff[d] for w,d in routes])
```

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EXHIBIT 7: DEFINING CONSTRAINTS FOR MIXED INTEGER LINEAR PROGRAMMING

```
#constraints
#outflow = inflow
model_t1 += lpSum([demand_decision_cno[w][d] for w,d in routes]) == lpSum([demand_decision_p2w_cno[p][w] for p,w in route_p2w_cno
model_t1 += lpSum([demand_decision_edible[w][d] for w,d in routes]) == lpSum([demand_decision_p2w_edible[p][w] for p,w in route_p
model_t1 += lpSum([demand_decision_vaho[w][d] for w,d in routes]) == lpSum([demand_decision_p2w_vaho[p][w] for p,w in route_p2w_v
```

Overall outflow of goods from warehouse to distributors = overall inflow of goods from plant to warehouse

```
for w in var_warehouse:
    #model_t1 += lpSum([demand_decision[w][d] for d in var_demand]) <= 100000*warehouse_open[w]
    model_t1 += lpSum(route_decision_cno[w][d] for d in var_demand) <= len(dist_w2d)*warehouse_open[w]
    model_t1 += lpSum(route_decision_edible[w][d] for d in var_demand) <= len(dist_w2d)*warehouse_open[w]
    model_t1 += lpSum(route_decision_vaho[w][d] for d in var_demand) <= len(dist_w2d)*warehouse_open[w]
```

Route decision = 1, if the warehouse is open

```
    model_t1 += lpSum(route_decision_cno[w][d]* distance_w2d[(w,d)] for d in var_demand) <= 500* len([distance_w2d[(w,d)] for d i
    model_t1 += lpSum(route_decision_edible[w][d]* distance_w2d[(w,d)] for d in var_demand) <= 500* len([distance_w2d[(w,d)] for
    model_t1 += lpSum(route_decision_vaho[w][d]* distance_w2d[(w,d)] for d in var_demand) <= 500* len([distance_w2d[(w,d)] for d
```

The average distance travelled to distributors from a warehouse must not exceed 500km or 2 service days

```
    model_t1 += lpSum(demand_decision_cno[w][d] for d in var_demand) == lpSum(demand_decision_p2w_cno[p][w] for p in plant_cno)
    model_t1 += lpSum(demand_decision_edible[w][d] for d in var_demand) == lpSum(demand_decision_p2w_edible[p][w] for p in plant_
    model_t1 += lpSum(demand_decision_vaho[w][d] for d in var_demand) == lpSum(demand_decision_p2w_vaho[p][w] for p in plant_vaho
```

Demand for each category of each warehouse is equal to the supply from plants of the respective category to the warehouse

```
    model_t1 += lpSum(route_decision_p2w_cno[p][w] for p in plant_cno) == warehouse_open[w]
    model_t1 += lpSum(route_decision_p2w_edible[p][w] for p in plant_edible) == warehouse_open[w]
    model_t1 += lpSum(route_decision_p2w_vaho[p][w] for p in plant_vaho) == warehouse_open[w]
```

Route from plant to warehouse active if warehouse open

Note: km = kilometre
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EXHIBIT 8: COST VERSUS SERVICE LEVEL GRAPHICAL REPRESENTATION

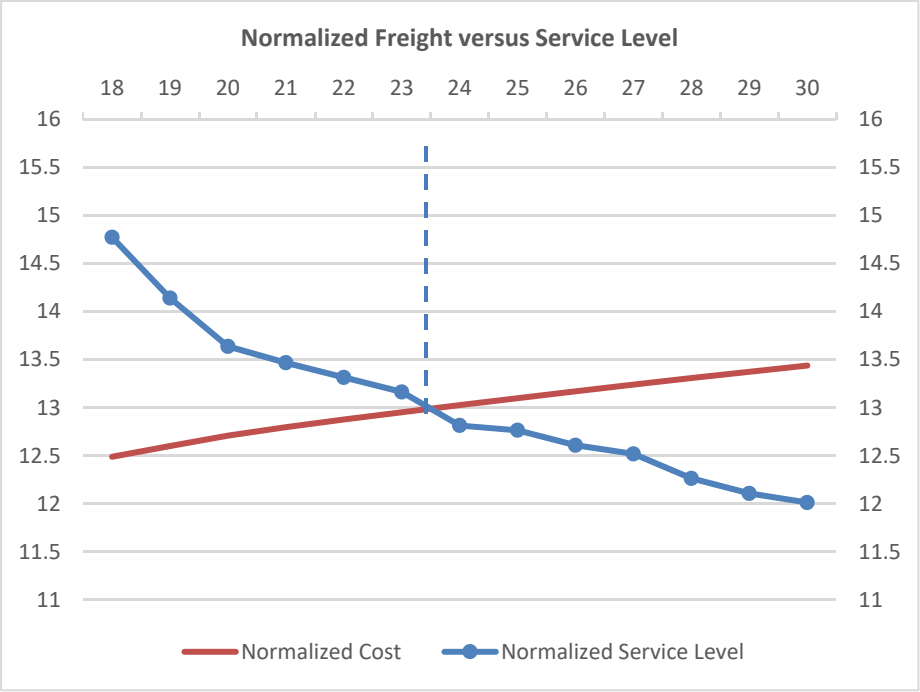

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EXHIBIT 9: INCREMENTAL IMPROVEMENT IN NORMALIZED FREIGHT AND SERVICE

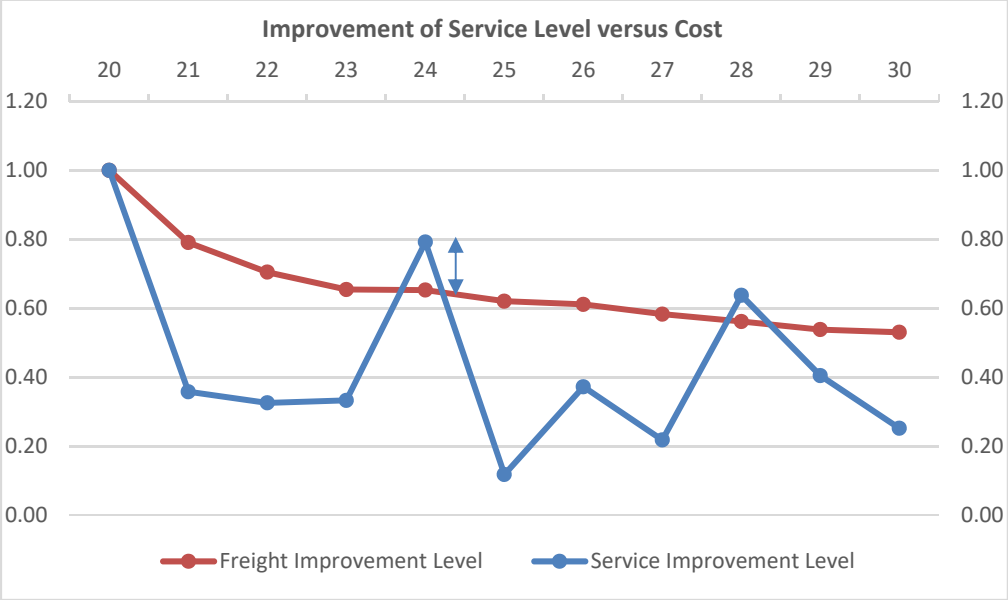

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EXHIBIT 10: DEFINING CUT-OFF CRITERIA AND CONSTRAINT FOR ALTERNATIVE ANALYSIS

```
#cut-off criteria for number of warehouses
    model_t1 +=lpSum(warehouse_open[w] for w in var_warehouse) <= z
#closing warehouses for alternative analysis
    model_t1 += warehouse_open['W9'] == 0
```

Defining the cut-off criteria with respect to total number of warehouses

Force shutting warehouses to understand the stability of the solution

Source: Created by the authors based on facts presented in the text.

EXHIBIT 11: SERVICE ANALYSIS OF 24 WAREHOUSES VERSUS CURRENT CONFIGURATION

	Demand <= 250 km	
Region	**Suggested**	**Current**
North	85%	73%
East	63%	63%
West	93%	79%
South	87%	78%

Note: km = kilometre
Source: Created by the authors based on facts presented in the text.

EXHIBIT 12: SERVICE ANALYSIS OF 22 WAREHOUSES VERSUS CURRENT CONFIGURATION

	Demand <= 250 km	
Region	**Suggested**	**Current**
North	75%	73%
East	65%	63%
West	93%	79%
South	83%	78%

Note: km = kilometre
Source: Created by the authors based on facts presented in the text.

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